ORDER-TO-CASH (O2C)

Complete Sales Cycle — SAP SD Module

SAP FIORI PROJECT REPORT

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Batch / Program: 2027 SAP FIORI (OE)

SAP UI5 1.120	SAP Fiori 3	OData v2	SAP HANA	S/4HANA

1. Project Title & Problem Statement

1.1 Project Title

Order-to-Cash (O2C) – Complete Sales Cycle | SAP Fiori Application (SAP SD Module)

1.2 Problem Statement

In large enterprises running SAP ERP / S/4HANA, the Order-to-Cash process spans multiple SAP SD transactions (VA11, VA21, VA01, VL01N, VF01, F-28) across different teams — sales, warehouse, finance, and accounts receivable. The inherent complexity causes the following challenges:

- **Fragmented Visibility:** Sales teams have no single view of where a customer order stands across the full cycle.
- **Manual Handoffs:** Teams rely on emails and phone calls to notify downstream steps, introducing delays and errors.
- **Data Silos:** Sales, logistics, and finance data exist in separate SAP modules with no unified dashboard.
- **Slow Dispute Resolution:** AR teams cannot quickly trace a payment dispute back to the originating sales order or delivery.
- **No Real-time KPIs:** Management lacks live insight into conversion rates, pending deliveries, and outstanding receivables.

Project Goal: Build a unified SAP Fiori application that digitizes and connects all six O2C steps into a single responsive interface, providing end-to-end traceability, live KPI dashboards, and one-click actions from inquiry creation through payment clearance.

2. Solution & Features

2.1 O2C Process Architecture

The application models the six canonical O2C steps as defined in SAP SD:

Step	Process	SAP T-Code	Description
Step 1	Customer Inquiry	VA11/VA12/VA13	Capture customer interest; record estimated

Step	Process	SAP T-Code	Description
			value and requirement description.
Step 2	Quotation	VA21/VA22/VA23	Generate formal price offer with validity period; convert accepted inquiry to quotation.
Step 3	Sales Order	VA01/VA02/VA03	Confirmed purchase commitment; line items with material, quantity, price, delivery date.
Step 4	Outbound Delivery	VL01N/VL02N	Pick, pack, and ship goods; Post Goods Issue (PGI) updates stock and O2C status.
Step 5	Billing / Invoice	VF01/VF02/VF03	Generate tax invoice (GST 18%); update accounts receivable in FI module.
Step 6	Payment Clearance	F-28 / FBL5N	Record inbound customer payment (NEFT/RTGS); clear open AR item; complete O2C cycle.

2.2 Application Features

- **Executive Dashboard:** Live KPI tiles — Inquiries, Quotations, Sales Orders, Pending Deliveries, Open Billing (INR), Paid This Month, Conversion Rate.
- **Process Navigator:** Visual 6-step O2C tile strip with arrows; one click navigates to the relevant module.
- **Inquiry Management:** Create (VA11), list, search, filter inquiries; convert Open inquiry to Quotation with a single button.
- **Quotation Management:** Create (VA21), list, view quotations; accept quotation converts it to a Sales Order automatically.
- **Sales Order CRUD:** Create (VA01) with full header and multi-line item entry, material lookup, GST auto-calculation; List/filter by status; Detail view with related-document trail.
- **Delivery Processing:** List all deliveries; Post Goods Issue per delivery or in batch; auto-updates Sales Order status to Delivered.
- **Billing Document:** Auto-generate invoice from delivered Sales Order; 18% GST calculation; tracks Open vs Paid billing per customer.

- **Payment Clearance:** Receive payment against open invoice via dialog (Method: NEFT/RTGS/IMPS/Cheque/Cash, UTR ref); clears AR and marks Sales Order as Paid.
- **Responsive Design:** SAP Fiori 3 Horizon theme; fully responsive across desktop, tablet, and phone using SAP UI5 device API.
- **i18n Ready:** All text externalised to i18n.properties; ready for EN/DE/JA translation.

2.3 O2C Status State Machine

Every Sales Order moves through a well-defined set of statuses:

Status	Meaning	Fiori State
Delivery Pending	SO created; awaiting delivery	Error (Red)
Delivered	PGI posted; goods left warehouse	Warning (Orange)
Billed	Invoice raised; awaiting payment	Warning (Orange)
Paid	Payment received and AR cleared	Success (Green)

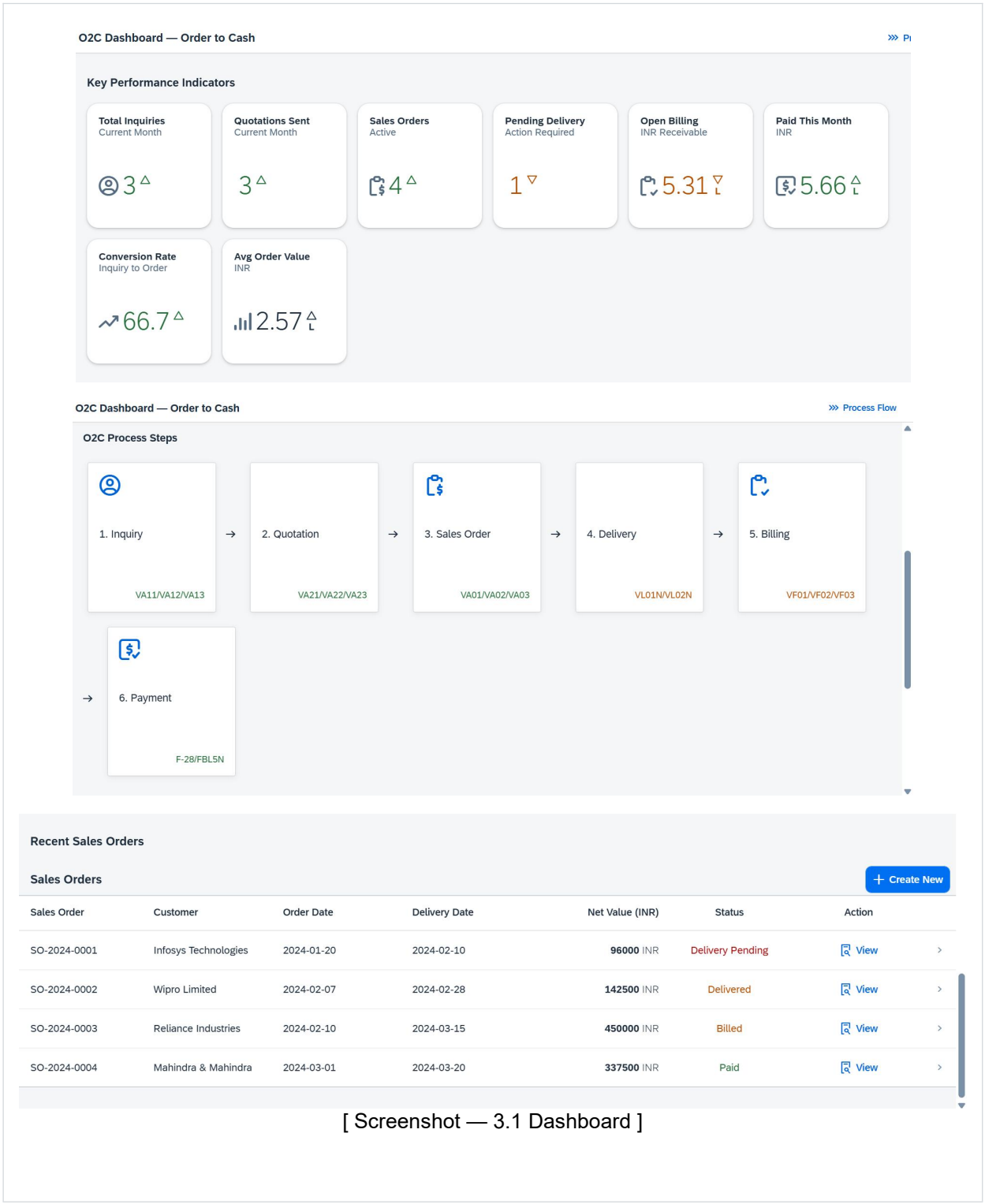
3. Application Screenshots (UI Wireframes)

The following describes the key screens of the application. Replace the placeholder boxes with actual screenshots from the running Fiori application before final submission.

3.1 Dashboard

The Dashboard is the application home screen. It aggregates all O2C KPIs in a tile grid (sap.f.GridList + GenericTile + NumericContent), displays the 6-step O2C process strip with navigation tiles, and shows the recent Sales Order table with live status badges.

- KPI Tile Row — 8 tiles showing Inquiries, Quotations, Sales Orders, Pending Deliveries, Open Billing, Paid This Month, Conversion Rate, Avg Order Value
- O2C Process Step Tiles — 6 steps with connecting arrows, each navigates to the relevant list screen
- Recent Sales Orders table with status-coloured ObjectStatus badges
- Create New button in the Sales Orders toolbar

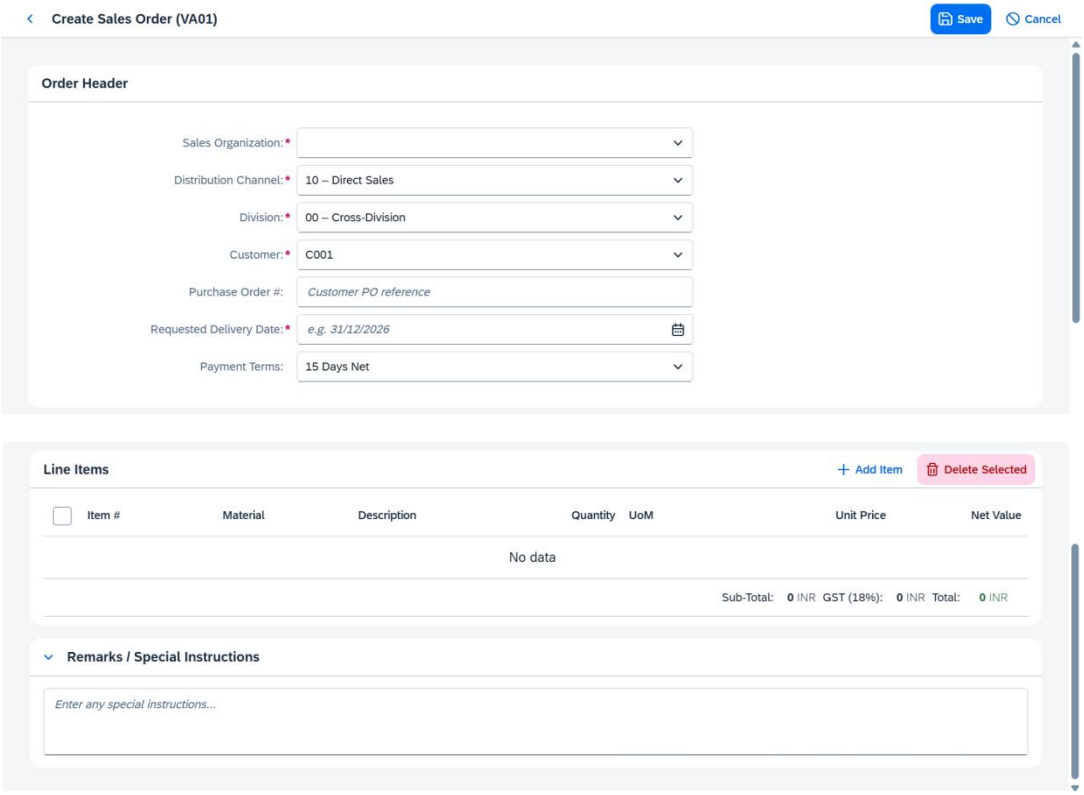


[Screenshot — 3.1 Dashboard]

3.2 Sales Order Create (VA01 equivalent)

A full create form replicating the SAP VA01 transaction. Selecting a material auto-fills description, UoM, and unit price. Changing quantity recalculates line value and grand total in real time.

- Header: Sales Org, Distribution Channel, Customer (select), PO Reference, Delivery Date, Payment Terms
- Line Item table: Add/Delete rows, Material Select with auto-fill, Qty input, Price auto-populated
- Footer: Sub-total, GST 18%, Grand Total (auto-calculated)
- Save / Cancel buttons in page header toolbar

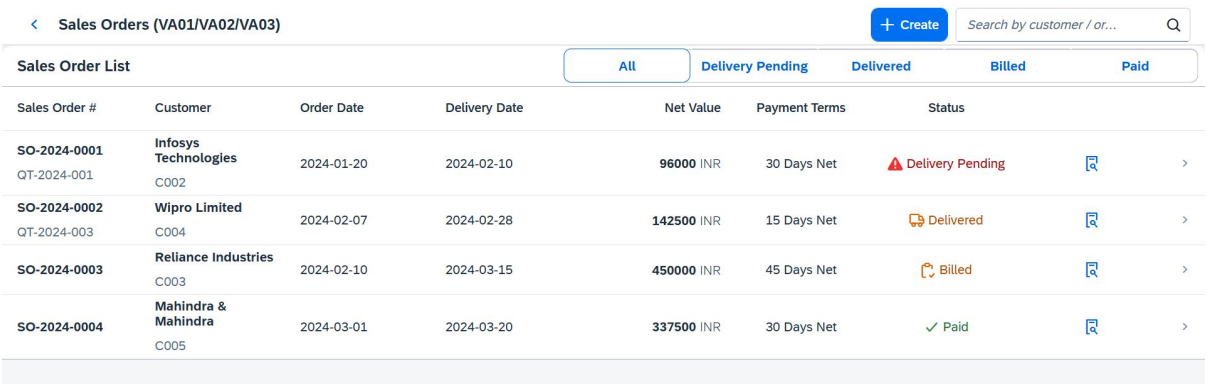


[Screenshot — 3.2 Sales Order Create (VA01 equivalent)]

3.3 Sales Order List (VA05 / VA03 equivalent)

The list screen for Sales Orders with client-side filtering and search using sap.ui.model.Filter and FilterOperator. Status filter is a SegmentedButton with 5 options.

- Filter bar with 5 status segments: All / Delivery Pending / Delivered / Billed / Paid
- Search field filtering by Customer Name or Sales Order ID
- Table with status-coloured ObjectStatus badges and navigation arrows
- Navigation to Sales Order Detail view on row click

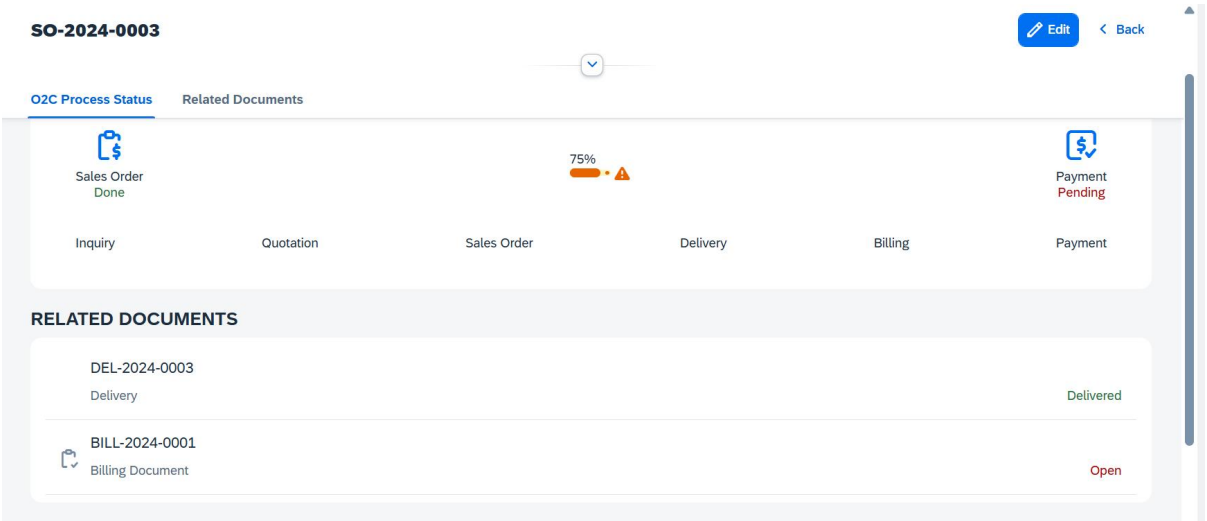


[Screenshot — 3.3 Sales Order List (VA05 / VA03 equivalent)]

3.4 Sales Order Detail (ObjectPage)

A sap.uxap.ObjectPageLayout detail view. The progress bar visualises how far the order has progressed through the O2C cycle. Related documents are listed with their current statuses.

- ObjectPage Dynamic Header: Customer, Order Date, Delivery Date, Payment Terms, Net Value
- Process Progress Bar: 25% (Delivery Pending) / 50% (Delivered) / 75% (Billed) / 100% (Paid)
- Related Documents section: linked Delivery, Billing Document, and Payment with statuses
- Edit button in header for future inline editing

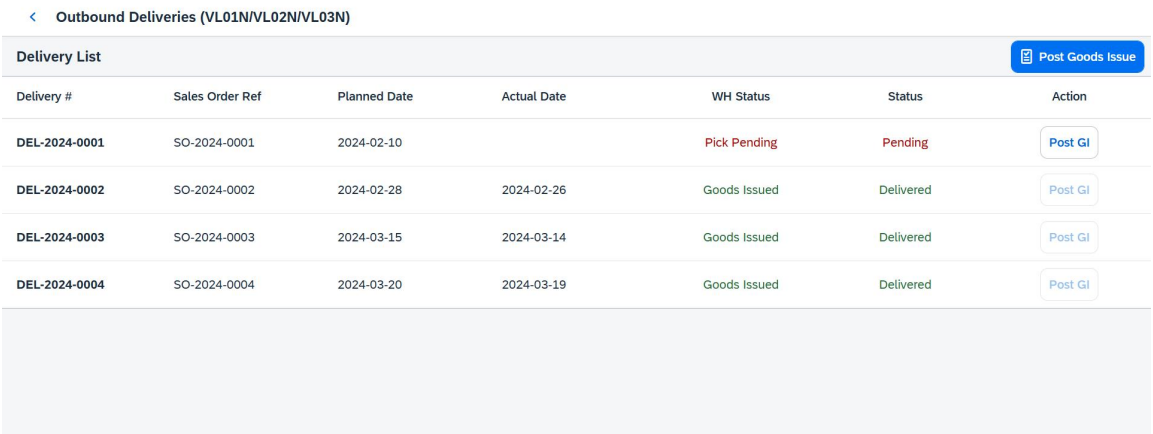


[Screenshot — 3.4 Sales Order Detail (ObjectPage)]

3.5 Delivery List (VL01N equivalent)

Posting a Goods Issue on this screen updates the Delivery to Delivered, updates the linked Sales Order status to Delivered, and decrements the KPI counter on the Dashboard.

- Table: Delivery ID, Sales Order Ref, Planned/Actual Dates, Warehouse Status, Delivery Status
- Per-row Post GI button — disabled for already-delivered rows
- Batch Post Goods Issue header button — posts all pending deliveries at once

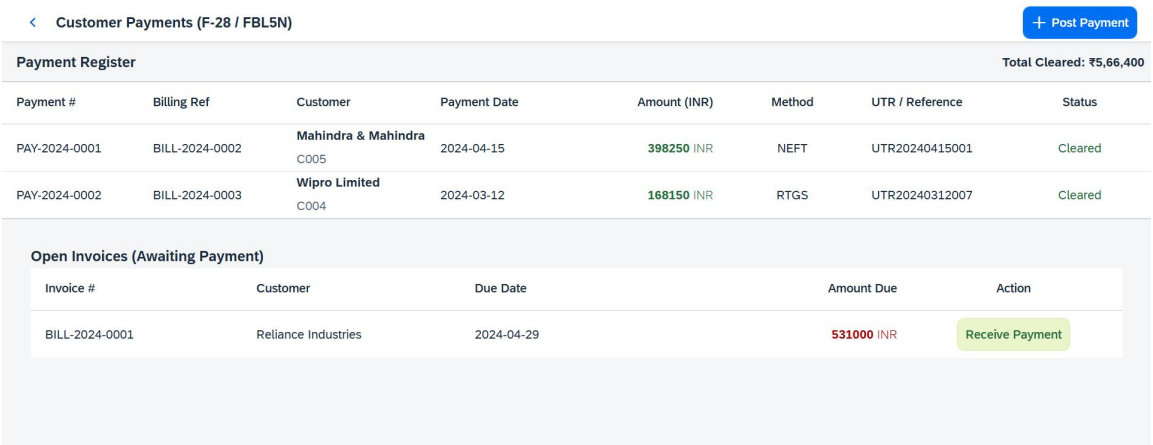


[Screenshot — 3.5 Delivery List (VL01N equivalent)]

3.6 Payment List (F-28 equivalent)

The payment screen presents a payment register (top) and open invoices (bottom). A dialog collects payment method and UTR reference before posting.

- Payment Register table: all cleared payments with UTR references and methods
- Open Invoices table: outstanding invoices with amount due in Error (red) state
- Receive Payment dialog: payment method selector (NEFT/RTGS/IMPS/Cheque/Cash) and UTR input
- Auto-clears AR and updates linked Sales Order status to Paid



[Screenshot — 3.6 Payment List (F-28 equivalent)]

4. Technology Stack

4.1 Frontend (SAP Fiori / UI5)

Component	Version / Variant	Purpose
SAP UI5	1.120.0 (LTS)	Core JavaScript framework — MVC, routing, data binding
SAP Fiori 3	Horizon theme	Design system — sap_horizon theme, responsive Fiori UX
sap.m	1.120.0	Mobile-first controls: Table, List, Page, Toolbar, Dialog
sap.f	1.120.0	Fiori controls: GridList, FlexibleColumnLayout
sap.uxap	1.120.0	ObjectPageLayout for detail views
sap.ui.layout	1.120.0	SimpleForm, HorizontalLayout, GridLayout
OData v2 Model	sap.ui.model.odata.v2	Two-way binding, batch requests, CRUD to SAP Gateway
JSONModel	sap.ui.model.json	Mock/offline data model for standalone demo
XML Views	Declarative	All views defined in XML for readability and performance
i18n	ResourceModel	Internationalisation — all labels externalised to i18n.properties

4.2 Backend (SAP Gateway / S/4HANA)

Component	Version	Role
SAP S/4HANA	2023 (on-premise/BTP)	ERP backend — SD, FI, MM modules
SAP Gateway	SAP_GWFND 2.0	OData service exposure layer
SAP OData Service	SD_O2C_SRV	Custom service — Inquiry, Quotation, SalesOrder, Delivery, Billing, Payment entity sets
SAP HANA DB	2.0 SPS06	Persistence layer for all O2C transaction data
ABAP CDS Views	DDIC-based	Data models powering OData entity types
SAP Fiori Launchpad	SAP_UI 7.58	Production deployment target — tile catalogues, target mappings

4.3 Development Toolchain

Tool	Usage
SAP Business Application Studio (BAS)	Cloud IDE — Fiori project templates, ESLint, Git integration
UI5 Tooling (ui5-cli)	Build, serve, lint — ui5 serve / ui5 build / ui5 add
SAP Fiori Tools	Page editor, guided development, deployment wizard
Git / GitHub	Version control — feature branches, pull requests, release tags
Node.js 18 LTS	Runtime for UI5 tooling and mock OData server
@sap/approuter	Cloud Foundry app router for Fiori deployment on BTP

5. Unique / Differentiating Points

1. End-to-End O2C in One App

Most SAP Fiori apps address a single transaction. This project covers all six O2C steps — Inquiry, Quotation, Sales Order, Delivery, Billing, and Payment — in a single coherent application with a shared data model and seamless navigation.

2. Live Status Propagation

Posting a Goods Issue on the Delivery screen automatically advances the linked Sales Order status, updates the Dashboard KPI tile, and reflects in the Payment screen — simulating real SAP document-flow event handling via a reactive JSONModel.

3. One-Click Document Conversion

Open Inquiry to Quotation to Sales Order conversion is a single button press, mirroring SAP VA01's Create with Reference flow without requiring the user to re-key data.

4. GST-Aware Billing Engine

The billing module automatically calculates 18% IGST on net order value, generates structured invoice data, and tracks open vs. paid receivables — directly relevant to the Indian GST compliance requirement for SAP customers.

5. SAP Fiori 3 Horizon Theme

Uses the latest sap_horizon theme with ObjectPageLayout, DynamicHeaderTitle, and GenericTile NumericContent — demonstrating modern Fiori 3 UX patterns beyond basic lists.

6. Fully Routed SPA

Every screen has a unique URL (hash-based routing via sap.m.routing.Router), enabling browser back-navigation, deep-linking, and Fiori Launchpad target-mapping compatibility.

7. Responsive and Offline-Ready

The sap.ui.Device model ensures tablet and phone adaptation. The JSON-based mock data model makes the app fully functional offline — critical for demo and presentation scenarios.

8. Reusable Formatter Module

All status-to-state mappings and icon lookups are centralised in model/formatter.js, following the SAP MVC best practice of separating presentation logic from view XML.

6. Future Improvements

1. SAP Gateway OData Integration

Replace the JSONModel mock with a live OData v2 service backed by ABAP CDS views and SAP Gateway. All CRUD operations will target the real SAP SD tables (VBAK, VBAP, LIKP, VBRK, BSID).

2. Credit Limit Check (FD32)

Integrate SAP FI customer credit exposure check during Sales Order creation. Block order creation if the customer's outstanding balance exceeds the credit limit, mirroring SAP's automatic credit control (T-code FD32 / OVA8).

3. PDF Invoice Generation

Generate a GSTIN-compliant PDF invoice using SmartForm / ADS (Adobe Document Services) and deliver it to the customer via email attachment directly from the Fiori app.

4. Multi-Currency Support

Extend the data model to handle multi-currency orders (USD, EUR, GBP) with live exchange rate conversion using SAP TCURR/TCURF tables.

5. SAP Analytics Cloud (SAC) Integration

Embed SAC Story dashboards within the Fiori ObjectPage to replace the static KPI tiles with interactive charts — AR aging analysis, sales trend, and on-time delivery rate.

6. Workflow / Approval Engine

Add SAP Business Workflow / BTP Process Automation steps for quotation approval (manager sign-off) and credit limit override, creating an auditable digital approval trail.

7. Batch Billing (VF04 equivalent)

Implement a selection screen to mass-create billing documents from multiple delivered Sales Orders in a single action, replicating the VF04 billing due list.

8. SAP Fiori Launchpad Deployment

Package the app as an SAPUI5 ABAP Repository (BSP) and deploy to SAP Fiori Launchpad with tile catalogues, target mappings, and role-based access control (PFCG).

7. Project File Structure

```
O2C_SAP_Fiori_Project/
  webapp/
    index.html                      Entry point (UI5 bootstrap)
    Component.js                    UIComponent — app init, routing start
    manifest.json                   App descriptor (routing, models, libs)
    controller/
      App.controller.js             Root — sets O2C JSONModel on component
      Dashboard.controller.js       KPI tiles, process navigation
      InquiryList.controller.js     VA13 list, search, convert to Quotation
      InquiryCreate.controller.js   VA11 create form
      QuotationList.controller.js   VA23 list, convert to Sales Order
      QuotationCreate.controller.js  VA21 create form
      SalesOrderList.controller.js   VA05 list, status filter, search
      SalesOrderCreate.controller.js VA01 create with line items
      SalesOrderDetail.controller.js VA03 detail, process progress
      DeliveryList.controller.js    VL03N list, Post GI action
      BillingList.controller.js     VF03 list, create invoice
      PaymentList.controller.js     FBL5N list, F-28 post payment
    view/
      App.view.xml                  Shell + App navigation container
      Dashboard.view.xml            KPI grid, process tiles, SO table
      InquiryList.view.xml
      InquiryCreate.view.xml
      QuotationList.view.xml
      QuotationCreate.view.xml
      SalesOrderList.view.xml
      SalesOrderCreate.view.xml
      SalesOrderDetail.view.xml     ObjectPageLayout
      DeliveryList.view.xml
      BillingList.view.xml
      PaymentList.view.xml
    model/
      models.js                     Device model + O2C mock data factory
      formatter.js                  Status, ValueState, icon, bool helpers
    i18n/
      i18n.properties               120 label keys (EN)
    css/
      style.css                     Custom Fiori overrides
  .gitignore                        Git ignore rules
  LICENSE                           MIT license file
  package.json                      npm scripts (start, build, lint)
  ui5.yaml                          UI5 tooling config (SAPUI5 1.120)
  .eslintrc.json                    ESLint rules
  README.md                         Setup & feature guide
  documentation/
    O2C_Project_Documentation.pdf   This document
```

8. SAP Transaction Code Mapping

The table below maps each Fiori screen and action to the equivalent SAP GUI transaction code and the underlying SAP SD/FI database tables affected.

Fiori Screen	T-Code	DB Tables	Function
Inquiry List	VA13	VBAK (type IN), VBAP	Display / Search Inquiries
Inquiry Create	VA11	VBAK, VBAP, VBKD	Create Customer Inquiry
Quotation List	VA23	VBAK (type QT), VBAP	Display / Search Quotations
Quotation Create	VA21	VBAK, VBAP, VBKD	Create Quotation with/without reference
Sales Order List	VA05	VBAK (type OR), VBAP	List Sales Orders by criteria
Sales Order Create	VA01	VBAK, VBAP, VBKD, VBUK	Create Standard Order (type OR)
Sales Order Detail	VA03	VBAK, VBAP, VBFA	Display Sales Order + document flow
Delivery List	VL06O	LIKP, LIPS	Outbound Delivery Monitor
Post Goods Issue	VL02N	LIKP, MKPF, MSEG	Post Goods Issue on Delivery
Billing List	VF05	VBRK, VBRP	List Billing Documents
Create Invoice	VF01	VBRK, VBRP, BSID	Create Billing Document from Delivery
Payment List	FBL5N	BSID, BSAD	Customer Line Item Display
Post Payment	F-28	BKPF, BSEG, BSAD	Post Incoming Customer Payment